Coherent Algeraic Sheaves

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Contents

Chapter 1

Sheaves

1.1 Operations on Sheaves

1.1.1 Definition of a sheaf

Let X be a topological space. A *sheaf of abelian groups* on X (or simply a *sheaf*) consists of:

(a) A *function* $x \to \mathcal{F}_x$ *sending every* $x \in X$ to an abelian group $\mathcal{F}_x$,

(b) A topology on the set $\mathcal{F}$, the sum of the sets $\mathcal{F}_x$.

If f is an element of $\mathcal{F}_x$, we set $\pi(f) = x$; the map π is called the projection of $\mathcal{F}$ onto X; the subset of $\mathcal{F} \times \mathcal{F}$ formed by the pairs (f, g) such that $\pi(f) = \pi(g)$ is denoted $\mathcal{F} + \mathcal{F}$.

With these definitions in place, we can state the two axioms to which the data (a) and (b) are subject:

(I) For all $f \in \mathcal{F}$, there exist open neighborhoods V of f and U of $\pi(f)$ such that the restriction of π at V is a homeomorphism of V onto U. (In other words, if *pi* is a local homeomorphism)

(II) The mapping $f \to -f$ is a continuous mapping of $\mathcal{F}$ into $\mathcal{F}$, and the mapping $(f, g) \to f + g$ is a continuous mapping of $\mathcal{F} + \mathcal{F}$ into $\mathcal{F}$.

It will be observed that, even if X is separated (which we do not assume), $\mathcal{F}$ is not necessarily separated, which is illustrated by the example of germs of functions.

Example of a sheaf. For an abelian group G, let us set $\mathcal{F}_x = G$ for every $x \in X$; the set $\mathcal{F}$ can be identified with the product $X \times G$, and, if we equip is with the product topology of X by the discrete topology of G, we obtain a sheaf, called the *constant sheaf* isomorphic with G, often identified with G.

1.1.2 Sections of a sheaf

Let $\mathcal{F}$ be a sheaf on the space X, and let U be a subset of X. We define a *s*ection of $\mathcal{F}$ over U to be a continuous map $s : U \to \mathcal{F}$ such that $\pi \circ s$ is the identity mapping on U. So we have $s(x) \in \mathcal{F}_x$ for all $x \in U$. The set of sections of F over U is denoted by $\Gamma(U, \mathcal{F})$; axiom (II) implies that $\Gamma(U, \mathcal{F})$ is an abelian group. If $U \subset V$, and if s is a section over V, the restriction of s at U is a section over U; hence we have a homomorphism $\rho_U^V : \Gamma(V, \mathcal{F}) \to \Gamma(U, \mathcal{F})$.

If U is open on X, $s(U)$ is open in $\mathcal{F}$, and, U runs over a base of the topology of X, then $s(U)$ is runs over a base of the topology of $\mathcal{F}$; this is only another way of stating axiom (I).

Note another consequence of axiom (I): For every $f \in \mathcal{F}_x$, there exists a section s over a neighborhood of x such that $s(x) = f$, and two sections enjoying this property coincide on a neighborhood of x. In other words, $\mathcal{F}_x$ is an *inductive limit* of $\Gamma(U, \mathcal{F})$ following the filtering order of the neighborhoods of x.

1.1.3 Construction of sheaves

Suppose we are given, for every open $U \subset X$, an abelian group $\mathcal{F}_U$, and, for every pair of open $U \subset V$, a homomorphism $\varphi_U^V : \mathcal{F}_V \to \mathcal{F}_V$, satisfying the transivity condition $\varphi_U^V \circ \varphi_V^W = \varphi_U^W$ whenever $U \subset V \subset W$.

The collections $(\mathcal{F}_U, \varphi_U^V)$ allows us to define a sheaf in the following way:

(a) We set $\mathcal{F}_x = \lim \mathcal{F}_U$ (inductive limit with respect to the filtered ordering of open neighborhoods U of x). If x belongs to an open set U, we have a canonical morphism $\varphi_x^U : \mathcal{F}_U \to \mathcal{F}_x$.

(b) Let $t \in \mathcal{F}_U$, and denote by $[t, U]$ the set of $\varphi_x^U(t)$ for x running over U; we have $[t, U] \subset \mathcal{F}$, and we give $\mathcal{F}$ the topology generated by $[t, U]$. Thus, an element $f \in \mathcal{F}_x$ admits as a neighborhood basis in $\mathcal{F}$ the sets $[t, U]$ where $x \in U$ and $\varphi_x^U(t) = f$.

We immediately verify that the data (a) and (b) satisfy the axioms (I) and (II), in other words, that $\mathcal{F}$ is indeed a sheaf. We say that this is the sheaf *defined by the system* $(\mathcal{F}_U, \phi_U^V)$.

If $t \in \mathcal{F}_U$, the mapping $x \to \phi_x^U(t)$ is a section of $\mathcal{F}$ over U; hence we have a canonical morphism $\iota : \mathcal{F}_U \to \Gamma(U, \mathcal{F})$.

Proposition 1. $\iota : \mathcal{F}_U \to \Gamma(U, \mathcal{F})$ *is injective if and only if the following condition holds*:

If an element $t \in \mathcal{F}_U$ has an open covering U_i of U with $\varphi_{U_i}^U(t) = 0$ for all i, then $t = 0$.

If $t \in \mathcal{F}_U$ satisfies the previous condition, we have

$$\varphi_x^U(t) = \varphi_x^{U_i} \circ \varphi_{U_i}^U(t) = 0 \text{if} x \in U_i,$$

which means that $\iota(t) = 0$. Conversely, suppose that $\iota(t) = 0$, with $t \in \mathcal{F}_U$; since $\varphi_x^U(t) = 0$ for $x \in U$, there exists an open neighborhood $U(x)$ of x such that $\varphi_{U(x)}^U(t) = 0$, by the definition of an inductive limit. The sets $U(x)$ therefore form an open covering of U satisfying the condition stated above.